Lesson Objective

Find the slope of a line on a grid using properties of slope triangles.

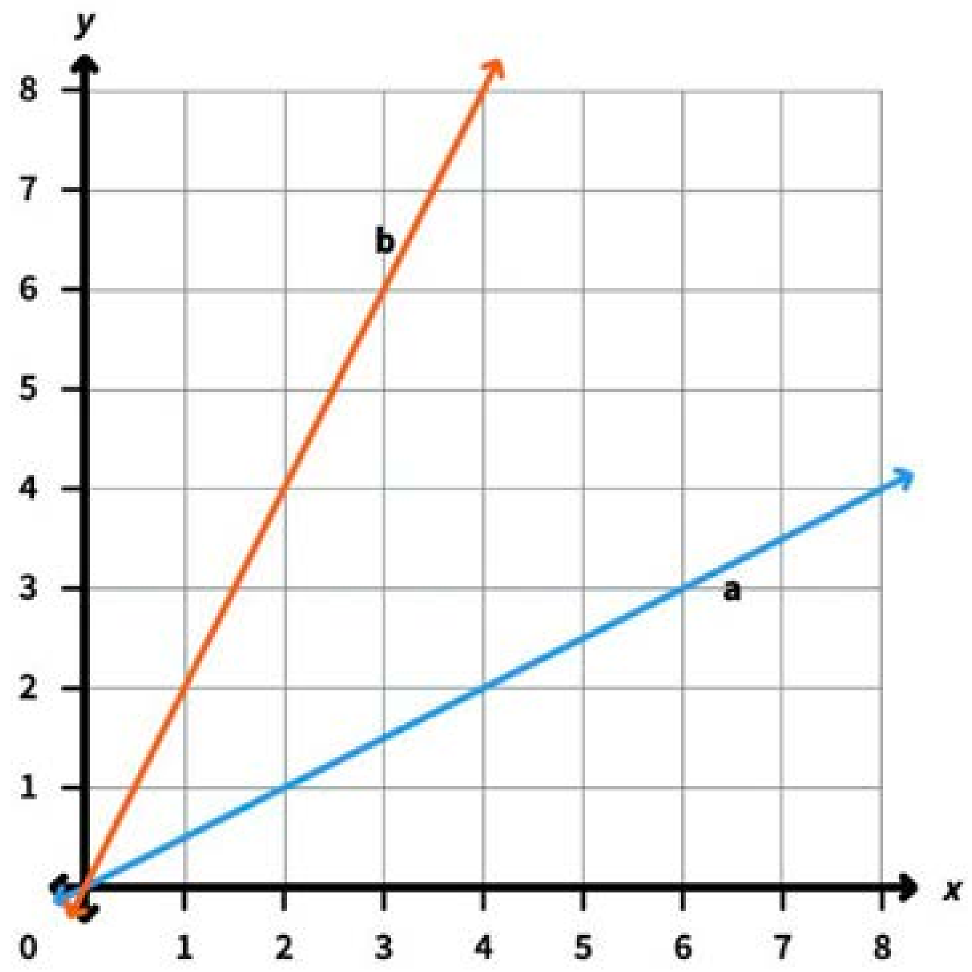
Materials

(S) Mini-Lesson Worksheet

PROBLEM	TEACHER GUIDANCE

1 **Identify which of two lines on a grid is steeper and connect steepness to the idea of slope as vertical distance divided by horizontal distance.**

Two lines are graphed on the grid below. Which line is steeper, line *a* or line *b*?



Answer: Line b is steeper.

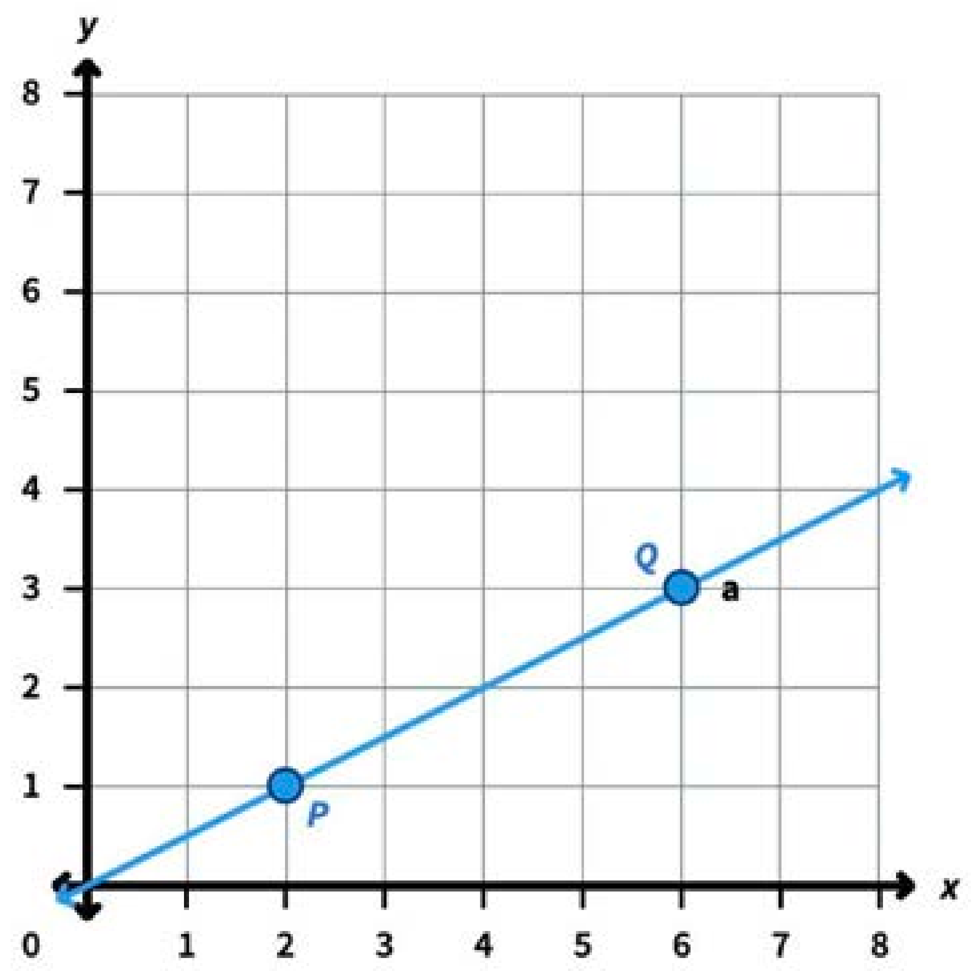
Ask students which line is steeper. If it helps, tell students to imagine each line is a slide — the one they would go down faster is steeper.

Tell students that slope is a number that describes the steepness of a line. Slope is the vertical distance divided by the horizontal distance between two points on the line. A steeper line has a greater slope.

Point out that in this lesson, students will learn how to find the slope of a line by picking two points on the grid and counting vertical and horizontal distances.

2 **Find the slope of line a by choosing two points at grid intersections and calculating the vertical distance divided by the horizontal distance.**

Find the slope of line *a*.



Answer: The slope of line a is 2/4, which equals 1/2.

Ask students to count the vertical distance from P to Q. Then ask students to count the horizontal distance from P to Q.

Have students record the slope as vertical distance divided by horizontal distance: 2/4. Ask students to simplify: 2/4 = 1/2.

Point out that a slope of 1/2 means the line goes up 1 unit for every 2 units across.

3 **Determine the slope of line b and compare slopes to describe the relative steepness of the two lines.**

Find the slope of line *b*. Then explain how the slopes of line *a* and line *b* confirm which line is steeper.

Answer: The slope of line a is 2/4, which equals 1/2.

Have students find the slope of line *b* using points C and D. As students are ready, have them count the vertical and horizontal distances and calculate the slope independently.

Ask students to compare the slopes of line *a* and line *b*. Ask which slope is greater and how that matches what they observed about steepness in Problem 1. Confirm that line *b* has a slope of 2, which is greater than line a's slope of 1/2, so line *b* is steeper.

Monitor For

- Do students correctly identify the vertical distance and horizontal distance between two points?
- Are students counting grid squares accurately, or are they off by one?
- Do students write slope as vertical distance divided by horizontal distance, not the reverse?
- Can students connect a larger slope value to a steeper line?

Instructional Moves

- If students confuse vertical and horizontal distances, have them trace the path: "Go straight up first, then straight across" to form the slope triangle.
- Remind students that slope is always vertical distance divided by horizontal distance — "rise over run" — not the other way around.
- If students struggle to count distances accurately, have them mark each grid square as they count from one point to the other.
- When comparing slopes, ask: "Which slope number is greater? That line is steeper. Why does that make sense?"

Reflection Questions

1. How did you use the grid to find the slope of a line?
Answer: I picked two points where the line crosses grid intersections, counted the vertical distance and the horizontal distance between them, and divided vertical distance by horizontal distance.
2. What does a slope of 1/2 tell you about how the line looks compared to a slope of 2?
Answer: A slope of 1/2 means the line only goes up 1 unit for every 2 units across, so it is less steep. A slope of 2 means the line goes up 2 units for every 1 unit across, so it is much steeper.
3. How does comparing slope values help you decide which line is steeper?
Answer: The line with the greater slope is steeper because it has a greater vertical distance for the same horizontal distance.